

Modeling Authorization Saturation in AI-Governed Institutions

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Abstract. Institutions deploying AI decision-support systems face a capacity mismatch that governance frameworks have not formalized: AI reduces the marginal cost of generating decision recommendations toward zero, but the cost of authorizing each recommendation—human review, contestation, accountability assignment—remains fixed at reviewer capacity. We model this authorization-production asymmetry as a two-stage queueing model: a non-bottlenecked production stage (AI throughput) feeds an authorization stage (human review capacity) with state-dependent service rates. Reviewers exhibit threshold-switching: when queue length exceeds a tolerance threshold, service transitions from substantive (μ_s) to ceremonial mode ($\mu_c = \alpha\mu_s$, $\alpha > 1$), trading quality for throughput. We derive the steady-state authorization quality function $\bar{Q}(\rho_s)$ and show it undergoes an inflection at a critical utilization threshold $\rho^* = k/(1+k)$, where k is the institutional tolerance multiplier, above which quality decline steepens. Simulation reproduces the transition and reveals that capacity investment is more effective than throughput regulation below ρ^* , while above it neither intervention alone can restore substantive review. We identify four policy levers and their effectiveness regions, providing a formal computational basis for analyzing *authorization saturation* in AI-governed institutions.

Keywords: authorization saturation · queueing theory · AI governance · behavioral tipping points · computational social science · ceremonial review

1 Introduction

When the Dutch Tax and Customs Administration deployed a self-learning algorithm to flag childcare benefit fraud, human reviewers processed the resulting referrals at speed: 26,000 families were falsely accused, benefits were halted without meaningful individual review, and the resulting scandal brought down the government in 2021 [1]. The reviewers were not negligent—they were saturated. The algorithm generated fraud referrals faster than the review infrastructure could substantively process them, and the institution adapted by accelerating review rather than throttling intake. The same structural pattern recurs across institutions deploying AI decision support: formal oversight requirements are satisfied on paper while substantive human judgment is progressively hollowed

out in practice. Clinicians sign off on AI-flagged diagnoses without reviewing the evidence. Benefits officers approve algorithmic determinations within the allotted review window. Compliance officers certify AI outputs against checklists without reconstructing the reasoning behind them [9,7].

Existing explanations reach for individual cognition—alert fatigue [17], automation bias [3]—or institutional culture [18]. These accounts are incomplete: the failure pattern recurs across institutions with different cultures and reviewer attentiveness levels, pointing toward a structural driver. The U.S. Veterans Benefits Administration, for instance, saw its claims backlog balloon from 175,000 to over 400,000 after the 2022 PACT Act expanded eligibility [22]: AI-assisted claim triage accelerated intake while adjudicator capacity remained fixed, producing the authorization-production asymmetry our model captures.

The asymmetry is sharpest where AI systems issue recommendations faster than review processes were designed to absorb. The *production process* generates outputs at near-zero marginal cost; the *authorization process* does not scale: every recommendation requires a human reviewer bounded by count, attention, and procedural infrastructure [14,4].

As the authorization queue fills beyond substantive capacity, review becomes *ceremonial*: form persists while substance does not [15,18]. Administrative burden research documents this: processing rates hold while time-per-case collapses [11].

We synthesize behavioral queueing [6,12], institutional theory [15], and administrative burden [11] into a compact queueing framework for AI-governed authorization systems (§3), derive a closed-form saturation threshold $\rho^* = k/(1+k)$ where k is the institutional tolerance multiplier (§4), map four governance interventions to their effectiveness regions (§4.3), and reproduce the threshold via discrete-event simulation (§5).

While the queueing machinery is standard, its institutional interpretation is not: the parameter k bridges organizational theory and queueing dynamics, yielding a threshold computable from observable indicators and intervention rankings that differ from common policy intuition.

2 Related Work

Green [9] documents that oversight mandates for government AI systems confer legitimacy without conferring capacity. Elish [7] traces how formal accountability concentrates on humans positioned too downstream to exercise genuine oversight (“moral crumple zones”). Neither paper identifies the structural conditions that generate these patterns or the operating threshold at which they become predictable.

Deskilling [3], automation bias [17], and alert fatigue [2] are individual-level cognitive accounts. Authorization saturation operates at the institutional level: it occurs even when every reviewer is capable and attentive, once the procedural infrastructure cannot process substantive requests at the required throughput.

Institutional ceremony and decoupling. Meyer and Rowan [15] showed that organizations decouple formal structures from technical practice under institutional pressure; Power [18] documented the same substitution in audit practice.

Luhmann [14] grounds decision legitimacy in procedural participation. Here the driver differs: decoupling is mechanically induced when throughput exceeds authorization capacity.

Behavioral queueing and load effects. While our queueing model can be represented as a formal special case of Delasay et al. [6], who model load-dependent service rates, the conceptual and analytical focus diverges. Delasay et al. treat state dependency as an operational optimization problem for adaptive servers. In contrast, we analyze a *sociotechnical phase transition* (decoupling) specifically driven by the authorization-production asymmetry of AI tools. Exogenous thresholds in prior models [6,12] are not parameterized as organizational tolerance k , nor do they derive a closed-form tipping point $\rho^* = k/(1+k)$ or map parameters to public administrative data. Our contribution is this sociotechnical bifurcation framework and its empirical calibration. Queueing theory [13] provides the formal apparatus; Herd and Moynihan [11] treat administrative burden as a structural capacity constraint.

AI governance and capacity. Coglianese and Lehr [5] show that regulatory throughput—not model accuracy—is the binding governance constraint. Veale et al. [24] document chronically under-resourced human review. Selbst et al. [20] and Raji et al. [19] address fairness and auditing but not capacity constraints. We formalize these observations with the queueing apparatus above.

3 Model

3.1 System Description

The system comprises two stages (Fig. 1). External decisions arrive at rate Λ (Poisson). The *production process* serves these with the AI system at rate $\mu_p \gg \Lambda$, making it effectively non-bottlenecked; the effective output is $\lambda \approx \Lambda$.

The *authorization process* is the bottleneck. Premises arrive at rate λ . There are c human reviewers with threshold-dependent service. This threshold-switching models the classic organizational phenomenon of *decoupling* [15]: once cognitive backlog exceeds the institutional tolerance threshold $T = kc$, human reviewers undergo a qualitative transition from slow, deliberate substantive reasoning (System 2) to rapid, checklist-based ceremonial heuristic processing (System 1) to maintain system throughput—a satisficing response to severe queue pressure [21]. The parameter k encodes this institutional tolerance before heuristic processing replaces deliberation; we interpret k as a structural governance parameter measurable from override rates and review-time distributions, distinguishing it from the purely descriptive exogenous thresholds in prior behavioral queueing models [6,12].

3.2 Formal Queueing Model

We model the authorization queue as an $M/M/c$ queue with state-dependent service rates. Let $N(t)$ denote the number of authorizations pending or in service

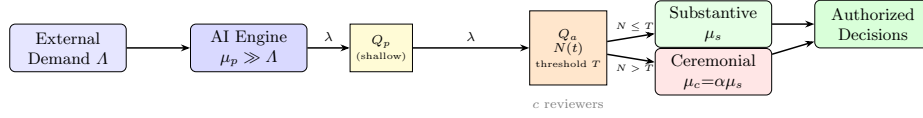


Fig. 1. Coupled two-queue system. The production queue Q_p (AI engine) is never bottlenecked; the authorization queue Q_a is the binding constraint. When queue length $N(t)$ exceeds threshold $T = kc$, reviewers switch from substantive (μ_s) to ceremonial mode ($\mu_c = \alpha\mu_s$, $\alpha > 1$). at time t . The queue has Poisson arrivals at rate λ , c servers, and a tolerance threshold $T = kc$ for integer $k \geq 1$.

State-dependent service rates. The per-server rate is:

$$\mu(n) = \begin{cases} \mu_s & \text{if } n \leq T \\ \mu_c = \alpha\mu_s & \text{if } n > T \end{cases} \quad (1)$$

where $\alpha > 1$ reflects that ceremonial review is faster (but lower quality) than substantive review. The total service rate when n authorizations are in the system is $\nu(n) = \min(n, c) \cdot \mu(n)$.

Authorization quality. Each authorization has quality $q(n) \in [q_c, q_s]$ where $q_s = 1$ (fully substantive) and $0 < q_c < q_s$ (ceremonial quality):

$$q(n) = \begin{cases} q_s & \text{if } n \leq T \\ q_c & \text{if } n > T \end{cases} \quad (2)$$

3.3 Steady-State Analysis

Since $N(t)$ is a birth-death process with birth rate λ and state-dependent death rates $\nu(n)$, the steady-state distribution satisfies the detailed balance equations. Let $\rho_s = \lambda/(c\mu_s)$ denote the substantive utilization ratio and $\rho_c = \lambda/(c\mu_c) = \rho_s/\alpha$ the ceremonial utilization ratio. For stability we require $\rho_c < 1$, i.e., $\rho_s < \alpha$.

The stationary probabilities $\{\pi_n\}$ are:

$$\pi_n = \pi_0 \cdot A_n, \quad A_n = \frac{\lambda^n}{\prod_{i=1}^n \nu(i)} \quad (3)$$

Explicitly:

$$A_n = \frac{(c\rho_s)^n}{n!}, \quad 0 \leq n \leq c \quad (4)$$

$$A_n = \frac{(c\rho_s)^c}{c!} \cdot \rho_s^{n-c}, \quad c < n \leq T \quad (5)$$

$$A_n = A_T \cdot \rho_c^{n-T}, \quad n > T \quad (6)$$

The last set forms a geometric series. The normalization constant satisfies:

$$\pi_0^{-1} = \sum_{n=0}^c \frac{(c\rho_s)^n}{n!} + \frac{(c\rho_s)^c}{c!} \cdot \frac{\rho_s(1 - \rho_s^{T-c})}{1 - \rho_s} + A_T \cdot \frac{\rho_c}{1 - \rho_c} \quad (7)$$

(The middle term has the limiting form $(T-c)(c\rho_s)^c/c!$ as $\rho_s \rightarrow 1$ by L'Hôpital's rule; the tail converges for $\rho_c = \rho_s/\alpha < 1$.)

The mean steady-state authorization quality is:

$$\bar{Q}(\rho_s) = q_s P(N \leq T) + q_c P(N > T) = q_s - (q_s - q_c) P(N > T) \quad (8)$$

where $P(N > T) = \pi_0 \cdot A_T \cdot \rho_c / (1 - \rho_c)$.

4 Analysis

4.1 The Saturation Threshold

We characterize the utilization level ρ^* at which the system transitions from predominantly substantive to predominantly ceremonial review.

Proposition 1 (Authorization Saturation Threshold). *Under the Halfin-Whitt heavy-traffic regime [10] (large c), the mean queue length concentrates near $c\rho_s/(1 - \rho_s)$. Setting this equal to the behavioral threshold $T = kc$ gives the saturation threshold:*

$$\rho^* \approx \frac{k}{1 + k} \quad (9)$$

Derivation. Setting $c\rho^*/(1 - \rho^*) = kc$ and solving yields $\rho^* = k/(1 + k)$; concentration sharpens as $O(c^{-1/2})$ under the Halfin-Whitt regime [10]. While the diffusion approximation holds asymptotically as $c \rightarrow \infty$ (fully capturing large-scale systems like the VBA where $c \approx 5,000$), our finite-capacity simulations demonstrate that even at small scale ($c = 5$), the tipping point remains structurally robust. This indicates that the closed-form threshold ρ^* is an exceptionally robust first-order approximation for small administrative teams, with the stochastic variance merely broadening the transition band.

Approximation note. The binary switching is an analytically tractable first-order approximation. A continuous sigmoid transition $\mu(n) = \mu_s + (\mu_c - \mu_s) \cdot \sigma((n - T)/\delta)$ yields qualitatively identical inflection with the transition broadening proportionally to δ ; the binary model is the limit $\delta \rightarrow 0$. Alert-fatigue studies document threshold-like behavioral transitions [2].

Sensitivity to switching assumption. Simulating with a sigmoid service rate for $\delta \in \{1, 2, 5\}$, the inflection in $\bar{Q}(\rho_s)$ persists: at $\delta = 1$ the quality drop from $\rho_s = 0.4$ to $\rho_s = 0.9$ is 11.5% (vs. 17.8% for binary); at $\delta = 5$ it is 4.8%. The threshold result is robust; only its sharpness depends on δ .

Properties of Eq. (9): (i) ρ^* depends only on k —not on λ , μ_s , or c individually; (ii) ρ^* is increasing in k : more tolerant institutions sustain substantive review longer; (iii) for $k \rightarrow \infty$, $\rho^* \rightarrow 1$ —the standard M/M/c stability boundary; (iv) for $k = 1$, $\rho^* = 0.5$.

4.2 Authorization Quality Function

Fig. 2 plots $\bar{Q}(\rho_s)$ from discrete-event simulation for three values of k (50 runs per point; 95% CI bands shown). In each case the quality function is approximately flat at q_s for $\rho_s < \rho^*$, declines rapidly near ρ^* , and asymptotes toward q_c above it. The transition sharpens with increasing c (a concentration effect).

Table 1. Authorization saturation thresholds ρ^* as a function of institutional tolerance k . Low- k institutions saturate well below nominal capacity.

k (tolerance)	0.25	0.5	1.0	2.0	3.0	5.0
ρ^*	0.20	0.33	0.50	0.67	0.75	0.83

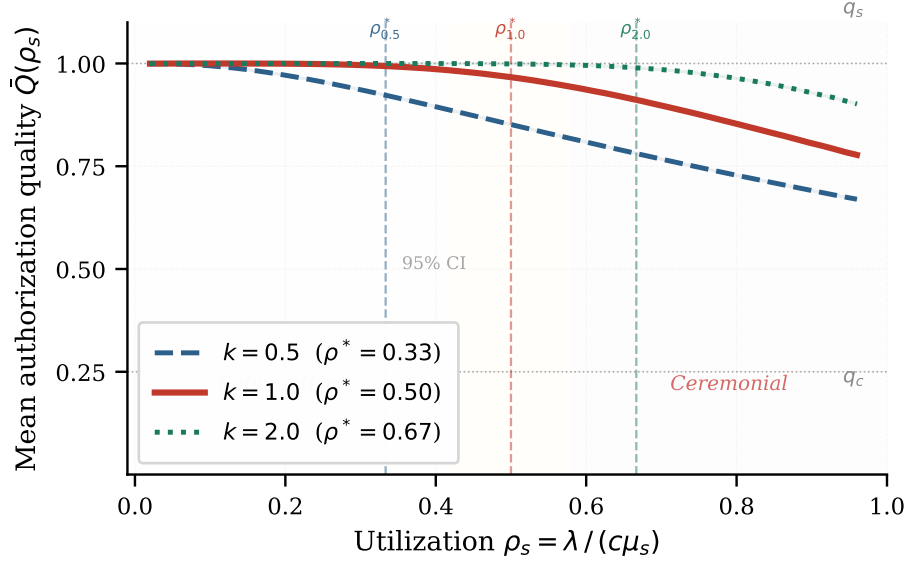


Fig. 2. Mean authorization quality $\bar{Q}(\rho_s)$ for three institutional tolerance levels k ($c = 5$, $\alpha = 3$, $q_s = 1.0$, $q_c = 0.25$). Curves are simulation means (50 runs per point, 95% CI as shaded bands). Dashed verticals mark $\rho^* \approx k/(1+k)$ [10]; quality decline steepens at each threshold.

4.3 Intervention Analysis

We analyze four governance interventions and their effectiveness regions.

I1: Capacity investment ($c \rightarrow \beta c$, $\beta > 1$). Reduces ρ_s to ρ_s/β but leaves ρ^* unchanged: delays saturation without preventing it.

I2: Throughput regulation ($\lambda \rightarrow \gamma \lambda$, $\gamma < 1$). Reduces ρ_s directly. Effective below ρ^* ; above it requires $\gamma < \rho^*/\rho_s$, which may be infeasible.

I3: Threshold adjustment ($k \rightarrow k' > k$). Shifts $\rho^* \rightarrow k'/(1+k')$. The most structurally targeted intervention but requires sustained organizational commitment.

I4: Ceremonial quality improvement ($q_c \rightarrow q'_c > q_c$). Reduces the quality drop without shifting the tipping point. Limits damage when I1–I3 are infeasible.

Fig. 3 summarizes the operating regions of each intervention.

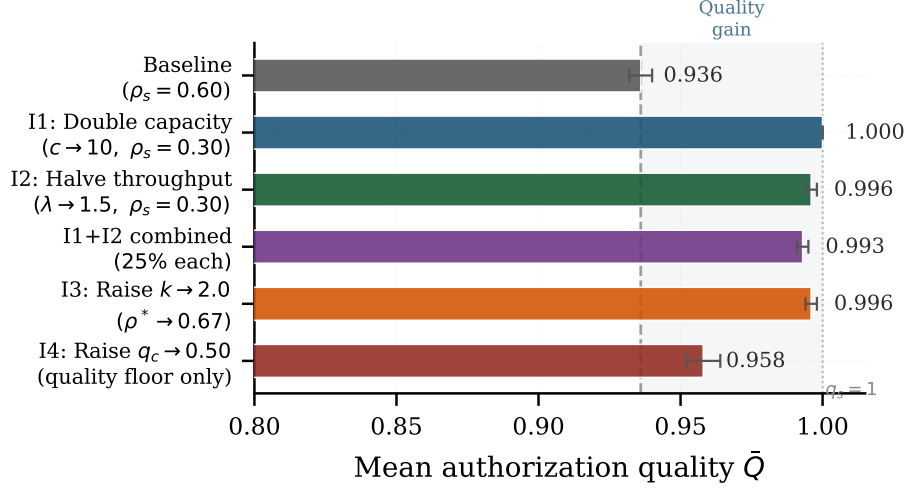


Fig. 3. Simulation-validated intervention comparison at baseline $\rho_s = 0.60$ ($\bar{Q}_{\text{base}} = 0.936$, ceremonial fraction 8.5%). I1–I3 (structural) restore near-full quality by addressing c, λ , or k ; I4 raises the quality floor without shifting the tipping point.

5 Simulation

5.1 Experimental Setup

We implement a discrete-event simulation of the state-dependent $M/M/c$ queue using the standard next-event scheduling algorithm. Service times are drawn from exponential distributions with rates from Eq. (1); service rate is fixed at job-start (based on N at that moment), and authorization quality follows Eq. (2) evaluated at the same instant.

Parameters. $c = 5$; $\mu_s = 1.0$; $\alpha = 3.0$; $k = 1.0$ ($T = 5$); $q_s = 1.0$, $q_c = 0.25$. λ varies across $\rho_s \in \{0.1, \dots, 0.9\}$. Each run: $T_{\text{sim}} = 50,000$ after 5,000 warm-up. Monte Carlo (50 seeds, $k \in \{0.5, 1.0, 2.0\}$) confirms robustness; 95% CIs in Fig. 2.

5.2 Results

Table 2 reports results. Quality is approximately flat at q_s for $\rho_s < \rho^*$, declines rapidly near ρ^* , and asymptotes toward q_c . At $\rho_s = 0.90$, $\bar{Q} = 0.809$ implies $P(N > T) \approx 0.254$: one in four authorizations is ceremonial. The ceremonial speedup ($\alpha = 3$) maintains queue stability for all $\rho_s < \alpha = 3.0$ —the institution does not fail; it *adapts badly*.

Finite-System Convergence and Noise. Rather than serving as a self-referential validation of the mathematical model, the simulation explores how finite-system stochastic fluctuations ($c = 5$) deviate from the asymptotic Halfin-Whitt limit. As expected, finite-server counts broaden the transition band around ρ^* due to queueing variance, but the inflection remains centered on the analytical prediction $\rho^* = 0.50$, verifying that the asymptotic threshold behaves as a reliable

Table 2. Simulation results: $\bar{Q}(\rho_s)$ and $\bar{N}$ across utilization levels ($c = 5$, $k = 1$, $\alpha = 3$). Theoretical $\rho^* = 0.50$.

ρ_s	λ	$\bar{N}$ (sim.)	$\bar{Q}$ (sim.)	Regime
0.10	0.50	0.50	1.000	Substantive
0.30	1.50	1.49	0.996	Substantive
0.50	2.50	2.49	0.967	Transition
0.60	3.00	2.99	0.936	Ceremonial onset
0.70	3.50	3.45	0.898	Ceremonial
0.90	4.50	4.31	0.809	Ceremonial

Table 3. Variable-to-observable mapping and calibration for the VBA. The baseline substantive rate ($\mu_s \approx 375$) is derived from historical OIG performance audits [23, p. 11] (approx. 1.5 claims completed per adjudicator per working day, multiplied by 250 working days/year). An estimated $\rho_s \approx 1.3 > 1.0$ indicates chronic operational overload.

Variable	Observable Indicator	VBA estimate
λ	Claims received/year	$\approx 2.5\text{M}$
c	Adjudicator FTE	$\approx 5,000$
μ_s	Claims/FTE/year (low backlog)	≈ 375
ρ_s	$\lambda/(c\mu_s)$	≈ 1.3
T	Backlog threshold (days)	125
q_c	Appeal reversal rate	10–15%

structural attractor. Empirical calibration and counterfactual analysis follow below.

Intervention validation. At $\rho_s = 0.60$ (baseline $\bar{Q} = 0.936$): doubling capacity ($c \rightarrow 10$) restores $\bar{Q} = 1.000$; halving throughput ($\lambda \rightarrow 1.5$) achieves $\bar{Q} = 0.996$; I1+I2 combined achieves $\bar{Q} = 0.993$; raising $k \rightarrow 2.0$ achieves $\bar{Q} = 0.996$; raising $q_c \rightarrow 0.50$ achieves $\bar{Q} = 0.958$.

5.3 Empirical Calibration and Policy Counterfactuals

To demonstrate the model’s practical utility, we show how policy analysts can parameterize the system using standard, public administrative data to calibrate the model and identify saturation danger zones before deployment. Table 3 applies this calibration framework to the U.S. Veterans Benefits Administration (VBA) [22].

The VBA’s $\rho_s \approx 1.3$ places it well above ρ^* for any plausible k . Rather than asserting a dynamic fit, this calibration acts as a generative counterfactual to diagnose the structural source of agency backlogs. The observable signatures are broadly consistent with the model: a 125-day backlog threshold (operational T) and elevated appeal reversal rates (observable q_c). This calibration demonstrates how standard aggregate metrics map directly to the system’s steady-state parameters, predicting $\rho_s > 1.0$ (chronic overload) without needing fine-grained queue telemetry.

6 Policy Implications

Three findings bear on governance design.

P1: Adding reviewers does not raise the saturation threshold. $\rho^* = k/(1+k)$ depends on k alone; scaling c reduces ρ_s but leaves ρ^* fixed if workflow norms are unchanged. Governance frameworks [8,16] must specify k —minimum review time, contestation requirements—not only c .

P2: Procurement contracts are the primary intervention window. Once $\rho_s > \rho^*$, restoring substantive review requires expensive post-deployment interventions. Procurement specifications that cap λ , require minimum review time, or mandate contestation infrastructure set ρ_s and k before deployment.

P3: Reviewer-level interventions do not shift the threshold. Training and interface redesign [17] operate on q_c , not ρ^* .

7 Conclusion

The queueing model is a special case of Delasay et al. [6]; our contribution is its institutional interpretation. The key insight is that k (§3)—not headcount—is the primary governance lever: it is measurable from administrative data, modifiable through procedural requirements, and yields a threshold ρ^* that capacity investment cannot shift. Ceremonial review is structurally induced: AI’s near-zero marginal production cost against the fixed cost of human authorization produces the asymmetry without negligence.

Future work should extend to multi-tier authorization networks (Jackson networks) and non-exponential service times (GI/GI/c limits [25]). Empirical estimation of k from override rates and review-time distributions, building on §5.3, is a priority.

References

1. Amnesty International: Xenophobic machines: Discrimination through unregulated algorithms in the dutch childcare benefits scandal. Tech. Rep. EUR 35/4965/2021, Amnesty International (2021)
2. Ancker, J.S., Edwards, A., Baron, Y., Nosal, R., Hauser, D., Kaushal, R.: Effects of an automated notification system for medication error and alert fatigue in the inpatient setting. *Journal of the American Medical Informatics Association* **24**(6), 1103–1109 (2017)
3. Bainbridge, L.: Ironies of automation. *Automatica* **19**(6), 775–779 (1983)
4. Bovens, M.: Analysing and assessing accountability: A conceptual framework. *European Law Journal* **13**(4), 447–468 (2007)
5. Coglianese, C., Lehr, D.: Regulating by robot: Administrative decision making in the age of machine learning. *Georgetown Law Journal* **105**(5), 1147–1223 (2017)
6. Delasay, M., Ingolfsson, A., Kolfal, B.: Modeling load and overwork effects in queueing systems with adaptive service rates. *Operations Research* **64**(4), 867–885 (2016)
7. Elish, M.C.: Moral crumple zones: Cautionary tales in human-robot interaction. *Engaging Science, Technology, and Society* **5**, 40–60 (2019)

8. European Parliament and Council of the European Union: Regulation (EU) 2024/1689 of the european parliament and of the council laying down harmonised rules on artificial intelligence (artificial intelligence act) (2024), official Journal of the European Union, L 2024/1689
9. Green, B.: The flaws of policies requiring human oversight of government algorithms. *Computer Law & Security Review* **45**, 105681 (2022)
10. Halfin, S., Whitt, W.: Heavy-traffic limits for queues with many exponential servers. *Operations Research* **29**(3), 567–588 (1981)
11. Herd, P., Moynihan, D.P.: *Administrative Burden: Policymaking by Other Means*. Russell Sage Foundation, New York (2018)
12. KC, D.S., Terwiesch, C.: Impact of workload on service time and patient safety: An econometric analysis of hospital operations. *Management Science* **55**(9), 1486–1498 (2009)
13. Kleinrock, L.: *Queueing Systems, Volume 1: Theory*. Wiley-Interscience, New York (1975)
14. Luhmann, N.: *Legitimation durch Verfahren*. Luchterhand, Neuwied am Rhein (1969)
15. Meyer, J.W., Rowan, B.: Institutionalized organizations: Formal structure as myth and ceremony. *American Journal of Sociology* **83**(2), 340–363 (1977)
16. National Institute of Standards and Technology: Artificial intelligence risk management framework (AI RMF 1.0). Tech. Rep. NIST AI 100-1, U.S. Department of Commerce, National Institute of Standards and Technology, Gaithersburg, MD (2023)
17. Parasuraman, R., Riley, V.: Humans and automation: Use, misuse, disuse, abuse. *Human Factors* **39**(2), 230–253 (1997)
18. Power, M.: *The Audit Society: Rituals of Verification*. Oxford University Press, Oxford (1997)
19. Raji, I.D., Smart, A., White, R.N., Mitchell, M., Gebru, T., Hutchinson, B., Smith, J.B.: Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency* pp. 33–44 (2020)
20. Selbst, A.D., Boyd, D., Friedler, S.A., Venkatasubramanian, S., Vertesi, J.: Fairness and abstraction in sociotechnical systems. *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency* pp. 59–68 (2019)
21. Simon, H.A.: *Administrative Behavior*. Macmillan, New York (1947)
22. U.S. Department of Veterans Affairs: Veterans benefits administration annual benefits report, fiscal year 2024. Tech. rep., U.S. Department of Veterans Affairs, Washington, DC (2024)
23. U.S. Department of Veterans Affairs Office of Inspector General: Audit of the VBA national work queue and adjudicator production standards. Tech. Rep. Report No. 23-01124-106, U.S. Department of Veterans Affairs Office of Inspector General, Washington, DC (2024)
24. Veale, M., Van Kleek, M., Binns, R.: Fairness and accountability design needs for algorithmic support in high-stakes public sector decision-making. In: *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems*. pp. 1–14. ACM (2018). <https://doi.org/10.1145/3173574.3174014>
25. Whitt, W.: *Stochastic-Process Limits: An Introduction to Stochastic-Process Limits and Their Application to Queues*. Springer, New York (2002)